

Luis Fernando Gaona Cárdenas

Power Electronics Engineer · Control Engineer · Researcher — Artificial Intelligence

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PROFESSIONAL PROFILE

Electronics Engineer with an M.Sc. and Ph.D. in Power Electronics, with research experience applying artificial intelligence to power electronics challenges: fault detection and classification using neural networks (up to 99.95% accuracy), implemented in real-time on FPGAs. Specialist in the design, simulation, and advanced control of power converters (DC-DC, AC-DC) for the automotive, industrial, and renewable energy sectors, with extensive experience in power PCB design, laboratory hardware validation, and model-based design cycles (MATLAB/Simulink, LTSpice). Author of indexed publications and speaker at international conferences (IEEE ROPEC).

CORE COMPETENCIES

- **Applied Artificial Intelligence:** Design, training, and validation of neural networks (MLP) for fault detection/classification and power quality disturbances; real-time FPGA implementation (Xilinx Basys 3, VHDL) with up to 99.95% accuracy; hybrid PID + AI control; optimization algorithms for control loops. Core development in MATLAB and embedded hardware; actively developing workflows in Python (PyTorch/TensorFlow).
- **Power Electronics Design:** DC-DC converters (Buck, Boost, Flyback, multiphase), AC-DC, inverters, Wide Bandgap (WBG) semiconductors (GaN/SiC), EMI mitigation.
- **Control & Modeling:** Digital control (PID, predictive, adaptive, AI-based), Model-Based Design (MBD), mixed-signal PCB design (DFM/DFA).
- **Software & Simulation:** MATLAB/Simulink, Altium Designer, LTSpice, PSIM, Vivado, Python (UI development), C++ (mathematical models).
- **Hardware Validation:** Oscilloscopes, spectrum analyzers, electronic loads, function generators.
- **Languages:** Spanish (Native), English (Professional working proficiency — reading, technical writing, and oral communication).

EDUCATION

Ph.D. in Electronic Engineering Sciences

Tecnológico Nacional de México — Celaya, México

August 2021 – July 2025

M.Sc. in Electronic Engineering

Tecnológico Nacional de México — Celaya, México

January 2018 – January 2020

B.Sc. in Electronics Engineering

Tecnológico Nacional de México — Celaya, México

August 2011 – November 2016

PUBLICATIONS AND PRESENTATIONS

- IEEE — PWM Strategy for Leakage Current in a Transformerless Multilevel Inverter. 2025 Mexican International Conference on Computer Science (ENC).
- IEEE — On the Analysis of Dual Phase Shift Modulation for Dual Active Bridge. 2025 International Conference on Electrical Drives and Power Electronics (EDPE).
- IEEE — FPGA Implementation of a Multi-Layer Perceptron. 2025 International Conference on Electrical Drives and Power Electronics (EDPE).
- IEEE — Online Multi-Layer Perceptron-Based Power Quality Monitoring for Electric Vehicles. 2024 IEEE 21st International Power Electronics and Motion Control Conference (PEMC).
- IEEE — Machine-Learning Based Fault Monitoring System for Electric Vehicle Onboard Chargers. IEEE Autumn Meeting on Power, Electronics and Computing (ROPEC), Ixtapa, México, 2022.
- MDPI — FPGA Implementation of Multilayer Perceptron for Real-Time Detection of Power Quality Disturbances for Electric Vehicle Charger (PQD for EVC). Information, 16(3), 180.
- MDPI — An Overview on Fault Management for Electric Vehicle Onboard Chargers. Electronics, 11(7), 1107.
- IJECE — Harvesting in Electric Vehicles: Combining Multiple Power Tracking and Fuel-Cells. Vol. 10, No. 5.
- CONCYS — Design and Simulation of a Parallel Bi-Directional Converter. National Conference on Circuits and Systems, Universidad Autónoma de San Luis Potosí, 2019.
- PE — Hardware Implementation of a FIR Filter with Distributed Arithmetic Using Serial and Parallel Architectures. National Conference on Renewable Energies, Instituto Tecnológico de Celaya, 2018.

CERTIFICATIONS

- Model-Based Design with Simulink — MathWorks, April 2025.
- VHDL and FPGA Development — Udemy, May 2025.
- Altium Designer Fundamentals — Altium Academy, May 2025.
- Power Electronics Specialization — Coursera, University of Colorado Boulder, December 2024.

KEY PROJECTS

Fault Detection for a Fault-Tolerant Rectifier (Ph.D. Thesis)

- Design of a three-phase PWM boost rectifier for electric vehicles.
- Use of a hybrid PID and Artificial Intelligence control.
- Artificial Intelligence algorithm for grid and switch fault detection.

Power Converter Design Software

- Development of mathematical models in Python.
- User interface (UI) design in Python.
- Implementation of optimization algorithms for control loops.

PROFESSIONAL EXPERIENCE

Power Electronics Research & Design Engineer (Ph.D. Project)

August 2021 – July 2025

Tecnológico Nacional de México — Guanajuato, Mexico

- Designed and optimized DC-DC and AC-DC power converters with a strong focus on high efficiency.
- Implemented advanced digital control strategies (PID, adaptive, and AI-based) in VHDL on FPGA platforms.
- Developed mixed-signal PCBs applying DFM/DFA principles using Altium Designer.
- Validated signal and power integrity (SI/PI) through comprehensive simulations in MATLAB and LTspice.
- Applied artificial neural networks for fault detection in AC-DC converters, achieving an accuracy rate above 95% (reaching up to 99.95% in the final published implementation).

Power Electronics Design Engineer

January 2020 – July 2021

PROSEELIA — Guanajuato, Mexico

- Designed custom power converters and their corresponding PCBs applying DFM/DFA guidelines in Altium Designer.
- Generated technical documentation and manufacturing deliverables (BOM, Gerber files) for production.
- Conducted electrical validation of prototypes and coordinated compliance testing for product certification.

Graduate Research Intern

August 2019 – December 2019

Ontario Tech University — Ontario, Canada

- Designed 2-layer and 4-layer PCBs for DC-DC converters utilized in electric mobility applications.
- Developed a multi-input converter for energy management across battery, supercapacitor, and solar panel sources.
- Validated system performance through PSIM simulations and rigorous laboratory hardware testing.
- Collaborated within an international research team and presented technical findings to the academic committee.

Product Engineer

November 2016 – January 2018

POLIGRAPACK — Guanajuato, Mexico

- Managed new product development (NPD) phases: defining specifications, design, prototyping, and validation testing.
- Generated complete technical documentation, including design specifications, process flowcharts, and user manuals.
- Developed and implemented Standard Operating Procedures (SOPs), reducing onboarding and training times for new personnel.
- Programmed and configured Programmable Logic Controllers (PLCs) for machinery automation.